

2. Analyze the two GenAI programs. The analysis must include:

- a) Name of two GenAI programs
 - I. ChatGPT and Claude
- b) How you accessed them (e.g., Google query, CoPilot on my computer)
 - I. App on computer (ChatGPT) and web browser (Claude)
- c) Prompt used. (You must use the same prompt for both GenAIs.)

"Use two GenAI's to generate C code to implement the following requirements (Note: You may use either ChatGPT or CoPilot as one of your choices, but not both. They are essentially the same GenAI.):

- The program will prioritize emails for a busy company CEO.
- You will use a MaxHeap as a means of implementing a priority queue. A priority queue is a queue where emails can shift towards the front of the queue based on a priority status.
- You must implement a MaxHeap using a list-based implementation. Then use that MaxHeap to handle all your email prioritizing for the CEO.
- You must create functions from scratch. Do not include pre-existing heap modules."

• Test File format:

Command	Description
EMAIL <sender category>, <subject line>, <date>	The emails in the CEO's Inbox should be placed in queue based on their sender category and date. The sender categories and priority to be read are as follows: • Boss – read first • Subordinate – read next • Peer – read next • ImportantPerson – read next • OtherPerson – read last If there is more than one from a sender, then the newest email (not the oldest) should be read first. I discovered this trick while a manager at Sprint. EMAIL is followed by space. The rest of the fields are delimited. Assume <sender category> is one of the five strings listed above. Assume <subject line> is a string which may contain spaces, but not commas Assume <date> is in the format: MM-DD-YYYY
NEXT	Next email for the CEO to read. Display the information on the terminal in the following format: Sender: Subject: Date:
READ	CEO has read email and dealt with it
COUNT	Display current count of unread emails

• Sample Test File:

```
EMAIL Peer,Can you help me on this?,12-01-2024
EMAIL OtherPerson,Try our product,12-19-2024
EMAIL Boss,Important,12-20-2024
EMAIL Subordinate,How do I handle this?,12-25-2024
EMAIL ImportantPerson,Health Insurance Enrollment,12-31-2024
EMAIL Boss,Never Mind,01-03-2025
COUNT
NEXT
READ
NEXT
READ
COUNT
```

• Your program should display this:

```
There are 6 emails to read.

Next email:
  Sender: Boss
  Subject: Never Mind
  Date: 01-03-2025

Next email:
  Sender: Boss
  Subject: Important
  Date: 12-20-2024

There are 4 emails to read.
```

- d) Generated code from both GenAIs.
- e) Analyze the code in terms of correctness, execution time, space complexity, and maintainability as explained in the “GenAI Code Comparison” lecture.
 - I. In terms of correctness, for Claude, when a date is replaced with nonsense, it still prints it. However for ChatGPT, it prints out that it's an invalid email line. For Claude, when the name of the sender that's not in the options, it assigns it to priority 0 and chooses the most recent date. For ChatGPT, it'll print out an error line.
 - II. For space complexity, Claude stores less variables than ChatGPT does, however they both have the same big O complexity. Claude also has a greater amount of while loops than ChatGPT does, increasing Claude's execution time.
 - III. Both Claude and ChatGPT both have a decent amount of comment lines that describe what the code is doing, however Claude has much more comment lines than ChatGPT.
- f) Select one to use as the basis for this assignment and justify why you picked it.
 - I. I've selected Claude for the assignment because the space complexity and the maintainability are better. I also like how the program still takes in the uncategorized name.
- g) Explain what you did to improve it in terms of correctness, execution time, space complexity, and maintainability.
 - I. Added edge cases for if dates are not correctly formatted and more comment lines.
- 3. Make the changes identified in 2g above.
- 4. Compile and debug your program.
- 5. Push both your source code, final cycle server executable file, and PDF of your GenAI analysis to your GitHub repository

RESULTS (for both):

There are 6 emails to read.

Next email:

Sender: Boss

Subject: Never Mind

Date: 01-03-2025

Next email:

Sender: Boss

Subject: Important

Date: 12-20-2024

There are 4 emails to read.